

2021-Ph-H1-Q19

Section: Particles and Waves

Topic: Refraction of Light

Which change occurs as light enters a denser medium?

Solution:

- Speed **decreases**
- Wavelength **decreases**
- Frequency stays **constant**

Answer: D

Guidance for Students:

Frequency doesn't change at a boundary. Speed and wavelength do. Light bends **towards the normal** when slowing down.

Revision Tip:

Use:

$$v = f\lambda$$

If $v \downarrow$, then $\lambda \downarrow$, but $f = \text{constant}$